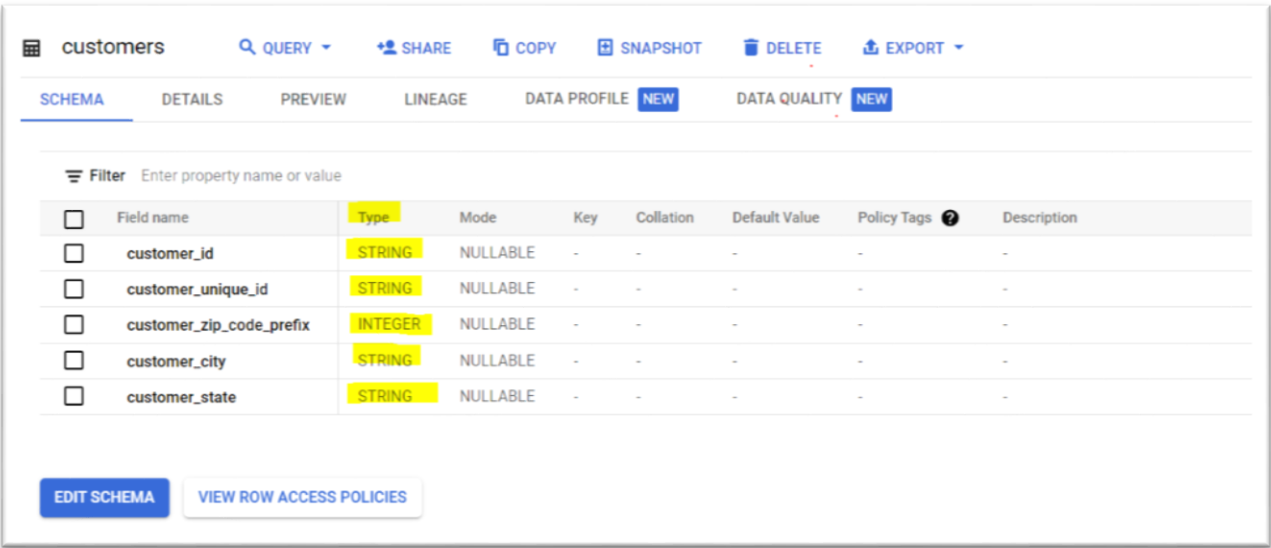


Question 1 :

Import the dataset and do usual exploratory analysis steps like checking the structure & characteristics of the dataset:

1A. Data type of all columns in the "customers" table.



customers								
SCHEMA DETAILS PREVIEW LINEAGE DATA PROFILE DATA QUALITY								
Filter Enter property name or value								
☐	Field name	Type	Mode	Key	Collation	Default Value	Policy Tags	Description
☐	customer_id	STRING	NULLABLE	-	-	-	-	-
☐	customer_unique_id	STRING	NULLABLE	-	-	-	-	-
☐	customer_zip_code_prefix	INTEGER	NULLABLE	-	-	-	-	-
☐	customer_city	STRING	NULLABLE	-	-	-	-	-
☐	customer_state	STRING	NULLABLE	-	-	-	-	-
EDIT SCHEMA VIEW ROW ACCESS POLICIES								

Summary :	Screenshot showing data-types of all columns in the customers table.
Observations:	Only the zip code for customer is of datatype 'int', rest all the ids (customer_id and customer_unique_id) are string type which gives freedom to create more unique and longer combinations for every customer. This will be very helpful as the number of customers increase with time !! Scalable design

1B. Get the time range between which the orders were placed.

Query	<pre>Select min(order_purchase_timestamp) as first_order_placed, max(order_purchase_timestamp) as last_order_placed from `target.orders`</pre>								
o/p	<table><tr><th>Row</th><th>first_order_placed</th><th>last_order_placed</th><th></th></tr><tr><td>1</td><td>2016-09-04 21:15:19 UTC</td><td>2018-10-17 17:30:18 UTC</td><td></td></tr></table>	Row	first_order_placed	last_order_placed		1	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC	
Row	first_order_placed	last_order_placed							
1	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC							
Summary	1. The first order was placed in 2016 and the last order was placed in 2018 thus the time range being 2016-2018 . 2. The orders table consists of all the information related to every order , thus the minimum of order purchase date will give us the date of first time an order was placed . 3. The maximum of the order purchase date gives us the time when last order was placed .								
Observations	1. UTC – Co-ordinated Universal Time has been used , so in the future if Target wants to expand to other continents , there will be no conflict while recording time for different zones – Uniformity in data								
Recommendation	1. Since the first order placed at around 9:15 at night and last order around 5:30 in evening – the site needs to be kept up and running during all times of the day								

1C. Count the Cities & States of customers who ordered during the given period.

Query	<pre>Select count(distinct c.customer_city) as Cities_of_customer, count(distinct c.customer_state) as States_of_customer from `target.customers` as c</pre>
	If the customer details of orders placed between the first and last orders is specifically asked then the below query would work <pre>Select count(distinct c.customer_city) as Cities_of_customer, count(distinct c.customer_state) as States_of_customer</pre>

	<pre>from `target.customers` as c left join `target.orders` as o on c.customer_id = o.customer_id where o.order_purchase_timestamp between "2016-09-04 21:15:19 UTC" and "2018-10-17 17:30:18 UTC"</pre>						
o/p	<table><tr><th>Row</th><th>Cities_of_customer</th><th>States_of_customer</th></tr><tr><td>1</td><td>4119</td><td>27</td></tr></table>	Row	Cities_of_customer	States_of_customer	1	4119	27
Row	Cities_of_customer	States_of_customer					
1	4119	27					
Summary	People from 4119 cities from the 27 states have placed an order with Target						
Observations	1. Brazil in all has 27 different states – i.e people from all states have ordered from Target 2. However , Brazil has 5,570 cities but people from (5570-4119 = 1451) 1451 cities have not purchased anything from target during 2016-2018 .						
Recommendation	1. Since people from 1451 cities have not purchased from Target , Target can advertise more in these cities and can find potential clients from there . 2. This will boost business across all states equally The below table shows the states that Target needs to focus for adding new customers thereby boosting business: <pre>Select c.customer_state, count(*) as No_orders_placed_from_state from `target.customers` as c group by 1 order by 2 desc</pre>						

customer_state	No_orders_placed_from_state
SP	41746
RJ	12852
MG	11635
RS	5466
PR	5045
SC	3637
BA	3380
DF	2140
ES	2033
GO	2020
PE	1652
CE	1336
PA	975
MT	907
MA	747
MS	715
PB	536
PI	495
RN	485
AL	413
SE	350
TO	280
RO	253
AM	148
AC	81
AP	68
RR	46

2.In-depth Exploration:

A.Is there a growing trend in the no. of orders placed over the past years?

Query	<pre>Select Extract(Year from order_purchase_timestamp)as Year_, Extract(Month from order_purchase_timestamp)as Month_, Count(order_id) as No_of_orders_placed from `target.orders` group by 1,2 order by 1,2</pre>																																															
o/p	<table><tr><th>Row</th><th>Year_ ▼</th><th>Month_ ▼</th><th>No_of_orders_placed ▼</th></tr><tr><td>1</td><td>2016</td><td>9</td><td>4</td></tr><tr><td>2</td><td>2016</td><td>10</td><td>324</td></tr><tr><td>3</td><td>2016</td><td>12</td><td>1</td></tr><tr><td>4</td><td>2017</td><td>1</td><td>800</td></tr><tr><td>5</td><td>2017</td><td>2</td><td>1780</td></tr><tr><td>6</td><td>2017</td><td>3</td><td>2682</td></tr><tr><td>7</td><td>2017</td><td>4</td><td>2404</td></tr><tr><td>8</td><td>2017</td><td>5</td><td>3700</td></tr><tr><td>9</td><td>2017</td><td>6</td><td>3245</td></tr><tr><td>10</td><td>2017</td><td>7</td><td>4026</td></tr></table>				Row	Year_ ▼	Month_ ▼	No_of_orders_placed ▼	1	2016	9	4	2	2016	10	324	3	2016	12	1	4	2017	1	800	5	2017	2	1780	6	2017	3	2682	7	2017	4	2404	8	2017	5	3700	9	2017	6	3245	10	2017	7	4026
Row	Year_ ▼	Month_ ▼	No_of_orders_placed ▼																																													
1	2016	9	4																																													
2	2016	10	324																																													
3	2016	12	1																																													
4	2017	1	800																																													
5	2017	2	1780																																													
6	2017	3	2682																																													
7	2017	4	2404																																													
8	2017	5	3700																																													
9	2017	6	3245																																													
10	2017	7	4026																																													
Summary	1. In the starting opening months (9,10,12- of year 2016) the sales were very low -as Target was establishing itself 2. Then the orders started growing exponentially in 2017 3. A peak in orders seen in November 2017 4. The peak dropped and came back to average at the end i.e Dec 2017 5. From 2018 starting only high amount of sales is seen for 3 months and then the number of orders stabilized between 6000-7000 orders per month 6. A steep drop is seen in the 9,10th month of 2018 7. No data is available for Nov 2016 – maybe store was closed for that month																																															
Observations	1. Orders started picking up exponentially in start 2017 till the end 2017 2. A peak can be seen in November may be because of Thanksgiving , Black Friday Sales																																															

	3. In 2018 ,Target established itself garnering avg orders between 6k-7k 4. A sudden drop is seen in Sep,Oct 2018 maybe due to presidential elections and a politician being stabbed in a rally – causing chaos around Brazil 5. Or maybe a competitor store opened here during that time
Recommendations	1. During festive seasons a) more discounts can be provided to encourage more buying b) Target should amp up the quantity of products and make sure the most sold products are in stock c) Care should be taken for faster delivery and managing more number of orders during high demands d) Brazilians spent on average 282 U.S. dollars per person during Christmas, out of which 117 U.S. dollars (around 41.5 percent) were spent on food. Keeping this in mind the inventory should be ordered 2. Brazil holds the title of the largest e-commerce market in Latin America and ranks as the 10th largest globally. With 150 million online shoppers, Black Friday's importance in the Brazilian economy keeps growing each year,a survey showed an astonishing \$3.3 billion USD spent on Thursday and Friday. Considering how big this opportunity is focus should be on gifts and gift items 2. Effort should be made to keep high stock in every price range – low to high catering to every class of customer . 3. A study can be made what is an average price range where most items are bought – more inventory and variety should be stocked for that section 4. Various schemes can be run like buybacks and exchange for high ticket items improving sales of expensive products .

B. Can we see some kind of monthly seasonality in terms of the no. of orders being placed?

Observation	1.A peak can be seen in November may be because of Thanksgiving , Black Friday Sales 2.December sales also were good due to Christmas 3.Then it suddenly drops after festivals cause a lot of money is spent during the festive season – here discounts ,vouchers etc can encourage more buying
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C. During what time of the day, do the Brazilian customers mostly place their orders? (Dawn, Morning, Afternoon or Night)

- 0-6 hrs : Dawn
- 7-12 hrs : Mornings
- 13-18 hrs : Afternoon
- 19-23 hrs : Night

Query	<pre>Select Case when Extract(hour from order_purchase_timestamp) between 0 and 6 then '0-6 hrs : Dawn' when Extract(hour from order_purchase_timestamp) between 7 and 12 then '7-12 hrs : Mornings' when Extract(hour from order_purchase_timestamp) between 13 and 18 then '13-18 hrs : Afternoon' when Extract(hour from order_purchase_timestamp) between 19 and 23 then '19-23 hrs : Night' else 'Unknown' End as order_time_of_day , Count(order_id) as total_orders_at_that_time from `target.orders` group by 1 order by Count(order_id) desc</pre>
-------	--

o/p

```
11 Select
12     Case
13         when Extract(hour from order_purchase_timestamp ) between 0 and 6 then '0-6 hrs : Dawn'
14         when Extract(hour from order_purchase_timestamp ) between 7 and 12 then '7-12 hrs : Mornings'
15         when Extract(hour from order_purchase_timestamp ) between 13 and 18 then '13-18 hrs : Afternoon'
16         when Extract(hour from order_purchase_timestamp ) between 19 and 23 then '19-23 hrs : Night'
17         else 'Unknown'
18     End as order_time_of_day ,
19     Count(order_id) as total_orders_at_that_time
20
21 from `target.orders`
22 group by 1
23 order by Count(order_id) desc
24
```

Query results

SAVE

JOB INFORMATION	RESULTS	CHART	PREVIEW	JSON	EXECUTION DETAILS	EXECUTION GRAPH
row	order_time_of_day	total_orders_at_that_time				
1	13-18 hrs : Afternoon	38135				
2	19-23 hrs : Night	28331				
3	7-12 hrs : Mornings	27733				
4	0-6 hrs : Dawn	5242				

Observations

1.Brazilians mostly make their purchases in the Afternoon - 13-18hrs time slot

2.2nd and 3rd highest – almost same – 19-23 :Night and 7-12hrs:Mornings slot

3. Very few orders made in the Dawn – 0-6hrs slot

Recommendation

1. Make sure the site is managed to handle more orders during this time

2. Tailor promotional campaigns and special offers to coincide with peak order times (Afternoon and Night). Consider offering time-specific discounts or promotions to incentivize orders during specific periods.

3. Schedule email campaigns, advertisements, and social media posts to reach customers during these high-engagement periods

III. Evolution of E-commerce orders in the Brazil region:

A. Get the month on month no. of orders placed in each state.

Query	<pre>Select Extract(Year from o.order_purchase_timestamp) as Year_, Extract(Month from o.order_purchase_timestamp) as month_, c.customer_state, Count(order_id)as no_of_orders from `target.orders` as o left join `target.customers` as c on o.customer_id=c.customer_id group by 1,2,3 order by 1,2 asc ----- If only month on month is required : Not Yearwise SELECT EXTRACT(MONTH FROM o.order_purchase_timestamp) AS month_, c.customer_state, COUNT(order_id) AS no_of_orders FROM `target.orders` AS o LEFT JOIN `target.customers` AS c ON o.customer_id = c.customer_id GROUP BY 1, 2 ORDER BY 1, 3 DESC;</pre>
-------	---

o/p	JOB INFORMATION RESULTS CHART PREVIEW JSON EXECUTION DETAILS					
	Row	Year_	month_	customer_state	no_of_orders	
	1	2016	9	RR	1	
	2	2016	9	RS	1	
	3	2016	9	SP	2	
	4	2016	10	AL	2	
	5	2016	10	BA	4	
	6	2016	10	CE	8	
	7	2016	10	DF	6	
	8	2016	10	ES	4	
	9	2016	10	GO	9	
	10	2016	10	MA	4	
	<u>If only month on month is required : Not Yearwise</u>					

	<table><tr><th>Row</th><th>month_</th><th>customer_state</th><th>no_of_orders</th></tr><tr><td>1</td><td>1</td><td>SP</td><td>3351</td></tr><tr><td>2</td><td>1</td><td>RJ</td><td>990</td></tr><tr><td>3</td><td>1</td><td>MG</td><td>971</td></tr><tr><td>4</td><td>1</td><td>PR</td><td>443</td></tr><tr><td>5</td><td>1</td><td>RS</td><td>427</td></tr><tr><td>6</td><td>1</td><td>SC</td><td>345</td></tr><tr><td>7</td><td>1</td><td>BA</td><td>264</td></tr><tr><td>8</td><td>1</td><td>GO</td><td>164</td></tr><tr><td>9</td><td>1</td><td>ES</td><td>159</td></tr><tr><td>10</td><td>1</td><td>DF</td><td>151</td></tr><tr><td>11</td><td>1</td><td>PE</td><td>113</td></tr><tr><td>12</td><td>1</td><td>CE</td><td>99</td></tr><tr><td>13</td><td>1</td><td>MT</td><td>96</td></tr><tr><td>14</td><td>1</td><td>PA</td><td>82</td></tr><tr><td>15</td><td>1</td><td>MS</td><td>71</td></tr><tr><td>16</td><td>1</td><td>MA</td><td>66</td></tr><tr><td>17</td><td>1</td><td>PI</td><td>55</td></tr><tr><td>18</td><td>1</td><td>RN</td><td>51</td></tr><tr><td>19</td><td>1</td><td>AL</td><td>39</td></tr><tr><td>20</td><td>1</td><td>PB</td><td>33</td></tr><tr><td>21</td><td>1</td><td>SE</td><td>24</td></tr><tr><td>22</td><td>1</td><td>RO</td><td>23</td></tr></table>	Row	month_	customer_state	no_of_orders	1	1	SP	3351	2	1	RJ	990	3	1	MG	971	4	1	PR	443	5	1	RS	427	6	1	SC	345	7	1	BA	264	8	1	GO	164	9	1	ES	159	10	1	DF	151	11	1	PE	113	12	1	CE	99	13	1	MT	96	14	1	PA	82	15	1	MS	71	16	1	MA	66	17	1	PI	55	18	1	RN	51	19	1	AL	39	20	1	PB	33	21	1	SE	24	22	1	RO	23
Row	month_	customer_state	no_of_orders																																																																																										
1	1	SP	3351																																																																																										
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20	1	PB	33																																																																																										
21	1	SE	24																																																																																										
22	1	RO	23																																																																																										
Observations	1. We are observing data from Sep 2016 till Oct 2018 month over month sorted yearwise: a) The states MG ,SP, RJ, RS , PR,SC ,BA in the same order have had a major share of orders coming from these states b) São Paulo (SP) and Rio de Janeiro (RJ) are highly urbanized states with large metropolitan areas which is why people have more urbanized view on online shopping c) Minas Gerais (MG) is known for its mining industry and is an important agricultural and industrial state. d) All these states are relatively more urbanized and population wise dense – hence more orders can be seen from there																																																																																												
Recommendation	1. It was observed an increasing trend in revenue and orders over time , yet during october and january sales are decreasing probably after Festival Sales. Introducing possible discount on not so running product can help sell more products during those low going months.																																																																																												

	2. Target needs to advertise more in the other less urbanized and sparsely populated areas to drive more sales 3. Target should focus on user experience in urbanized area as there is scope for more people buying online here by gaining popularity by word of mouth
--	--

B. How are the customers distributed across all the states?

Query	<pre>Select customer_state , Count(distinct customer_id) as customer_count from `target.customers` group by 1 order by 2 desc</pre>																																																																						
o/p	<table><tr><th colspan="2">JOB INFORMATION</th><th>RESULTS</th><th>CHART</th><th>PREVIEW</th></tr><tr><th>Row</th><th>customer_state</th><th>customer_count</th><th></th><th></th></tr><tr><td>1</td><td>SP</td><td>41746</td><td></td><td></td></tr><tr><td>2</td><td>RJ</td><td>12852</td><td></td><td></td></tr><tr><td>3</td><td>MG</td><td>11635</td><td></td><td></td></tr><tr><td>4</td><td>RS</td><td>5466</td><td></td><td></td></tr><tr><td>5</td><td>PR</td><td>5045</td><td></td><td></td></tr><tr><td>6</td><td>SC</td><td>3637</td><td></td><td></td></tr><tr><td>7</td><td>BA</td><td>3380</td><td></td><td></td></tr><tr><td>8</td><td>DF</td><td>2140</td><td></td><td></td></tr><tr><td>9</td><td>ES</td><td>2033</td><td></td><td></td></tr><tr><td>10</td><td>GO</td><td>2020</td><td></td><td></td></tr><tr><td>11</td><td>PE</td><td>1652</td><td></td><td></td></tr><tr><td>12</td><td>CE</td><td>1336</td><td></td><td></td></tr></table>	JOB INFORMATION		RESULTS	CHART	PREVIEW	Row	customer_state	customer_count			1	SP	41746			2	RJ	12852			3	MG	11635			4	RS	5466			5	PR	5045			6	SC	3637			7	BA	3380			8	DF	2140			9	ES	2033			10	GO	2020			11	PE	1652			12	CE	1336		
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11	PE	1652																																																																					
12	CE	1336																																																																					
Observations	1.The maximum customers are distributed SP,RJ,MG,RS,SC,BA,DF,ES,GO,PE in desc order 2. São Paulo (SP) is the economic powerhouse of Brazil, being the most economically developed and industrialized state and hence the majority of customers come from there 3.Rio de Janeiro (RJ) has a strong presence in the oil and gas industry, and it's known for its cultural significance and tourism - again people with lot of money																																																																						
Recommendation	1. The industrialization and urbanization of states is directly proportional to number of customers buying from Target 2. People being more educated in the urbanized states also lead to more online buying , hence if Target can appeal to these masses through advertisement , it can increase its customer base here 3. As the rural areas see less customers , Target should focus on ways through media that appeals to rural audience , mobile phones are available even in rural areas																																																																						

	4. So sending messages or advertisements through mobiles can lead to more sales from these areas 5. Increasing network in north brazil , having small towns can help increase the customer base. As north Brazil has the worlds largest river and most extesive rain forest, must be a good travel destination, introducing necessary survival/ camping/adventure products can help increase revenue and order from northern region . introducing new different more products from top selling categories can increase revenue more. 6. Discounts always drive sales
--	--

IV. Impact on Economy: Analyze the money movement by e-commerce by looking at order prices, freight and others.

A. Get the % increase in the cost of orders from year 2017 to 2018 (include months between Jan to Aug only).

Query	<pre> Select T.cost_2017, T.cost_2018, round((T.cost_2018 - T.cost_2017)/T.cost_2017 * 100) as percentage_increase from (SELECT SUM(CASE WHEN EXTRACT(YEAR FROM o.order_purchase_timestamp) = 2017 AND EXTRACT(MONTH FROM o.order_purchase_timestamp) BETWEEN 1 AND 8 THEN p.payment_value ELSE 0 END) AS cost_2017, SUM(CASE WHEN EXTRACT(YEAR FROM o.order_purchase_timestamp) = 2018 AND EXTRACT(MONTH FROM o.order_purchase_timestamp) BETWEEN 1 AND 8 THEN p.payment_value ELSE 0 END) AS cost_2018 </pre>
-------	--

```

FROM
    target.orders o
JOIN
    target.payments p
ON o.order_id = p.order_id
WHERE
    EXTRACT(YEAR FROM o.order_purchase_timestamp) IN (2017, 2018)
    AND EXTRACT(MONTH FROM o.order_purchase_timestamp) BETWEEN 1 AND 8)T

Select
    T.cost_2017,
    T.cost_2018,
    round((T.cost_2018 - T.cost_2017)/T.cost_2017 * 100) as percentage_increase
from

(SELECT
    SUM(CASE
        WHEN EXTRACT(YEAR FROM o.order_purchase_timestamp) = 2017
        AND EXTRACT(MONTH FROM o.order_purchase_timestamp) BETWEEN 1 AND 8
        THEN p.payment_value
        ELSE 0
    END) AS cost_2017,

    SUM(CASE
        WHEN EXTRACT(YEAR FROM o.order_purchase_timestamp) = 2018
        AND EXTRACT(MONTH FROM o.order_purchase_timestamp) BETWEEN 1 AND 8
        THEN p.payment_value
        ELSE 0
    END) AS cost_2018

FROM
    target.orders o
JOIN
    target.payments p
ON o.order_id = p.order_id
WHERE
    EXTRACT(YEAR FROM o.order_purchase_timestamp) IN (2017, 2018)
    AND EXTRACT(MONTH FROM o.order_purchase_timestamp) BETWEEN 1 AND 8)T

```

o/p

Row	cost_2017	cost_2018	percentage_increase
1	3669022.119999...	8694733.839999...	137.0

Observations

From yearly point of view from 2017 to 2018 , considering only the period between Jan to Aug a percentage increase of 137% is seen which is phenomenal

Recommendation	1. A study should be done to understand which items caused a significant amount of percentage increase in sales from 2017 to 2018 . Then care should be taken that, that section is always full and new varieties are added driving more sales 2. The sections which did not drive increase in sales should be kept near cash counter so that people know about those items also and are attracted to purchase them – again increasing the sales 3. During sales try to keep maximum discount on dead inventory /stock, schemes like buy1 get 1 etc can be run for the dead stock

B. Calculate the Total & Average value of order price for each state.

Query	<pre> Select c.customer_state, round(sum(p.payment_value),2) as total_order_price, round(avg(p.payment_value),2) as average_order_price from `target.customers` as c left join `target.orders` as o on c.customer_id=o.customer_id left join `target.payments` as p on o.order_id=p.order_id group by 1 order by 2 desc </pre>
-------	---

o/p	Row	customer_state	total_order_price	average_order_price
	1	SP	5998226.96	137.5
	2	RJ	2144379.69	158.53
	3	MG	1872257.26	154.71
	4	RS	890898.54	157.18
	5	PR	811156.38	154.15
	6	SC	623086.43	165.98
	7	BA	616645.82	170.82
	8	DF	355141.08	161.13
	9	GO	350092.31	165.76
	10	ES	325967.55	154.71
Observations	1. The highest total order price is between states SP,RJ ,MG , till Es in desc order 2. However, the highest average price is 248 which is from PB			
Recommendation	1. This shows that high priced items have buyers from PB i.e rich crowd lives there so expensive items can be pushed in these areas . 2. The highest total order price is from SP,RJ,MG etc but they have some of the lowest average price i.e low cost items are bought in these states , i.e mostly middle class and lower middle class buys from these states . 3. EMI based payment incentives in these states will boost sales. 4. Hence , more sales of less value can be easily shipped to these people if warehouses are located in these states			

C. Calculate the Total & Average value of order freight for each state.

Query	<pre>SELECT c.customer_state, round(SUM(oi.freight_value),2) AS total_freight, round(AVG(oi.freight_value),2) AS average_freight FROM target.orders o JOIN target.customers c ON o.customer_id = c.customer_id JOIN target.order_items oi ON o.order_id = oi.order_id GROUP BY 1 order by 2 desc</pre>																																															
o/p	<table><tr><th>Row</th><th>customer_state</th><th>total_freight</th><th>average_freight</th></tr><tr><td>1</td><td>SP</td><td>718723.07</td><td>15.15</td></tr><tr><td>2</td><td>RJ</td><td>305589.31</td><td>20.96</td></tr><tr><td>3</td><td>MG</td><td>270853.46</td><td>20.63</td></tr><tr><td>4</td><td>RS</td><td>135522.74</td><td>21.74</td></tr><tr><td>5</td><td>PR</td><td>117851.68</td><td>20.53</td></tr><tr><td>6</td><td>BA</td><td>100156.68</td><td>26.36</td></tr><tr><td>7</td><td>SC</td><td>89660.26</td><td>21.47</td></tr><tr><td>8</td><td>PE</td><td>59449.66</td><td>32.92</td></tr><tr><td>9</td><td>GO</td><td>53114.98</td><td>22.77</td></tr><tr><td>10</td><td>DF</td><td>50625.5</td><td>21.04</td></tr></table>				Row	customer_state	total_freight	average_freight	1	SP	718723.07	15.15	2	RJ	305589.31	20.96	3	MG	270853.46	20.63	4	RS	135522.74	21.74	5	PR	117851.68	20.53	6	BA	100156.68	26.36	7	SC	89660.26	21.47	8	PE	59449.66	32.92	9	GO	53114.98	22.77	10	DF	50625.5	21.04
Row	customer_state	total_freight	average_freight																																													
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8	PE	59449.66	32.92																																													
9	GO	53114.98	22.77																																													
10	DF	50625.5	21.04																																													
Observations	1.(SP) has the highest total freight value, indicating that it might be a region with high order volume or greater distances to cover in terms of logistics. 2.PB,RR have the highest average freight value. This could be due to specific challenges in shipping to this region, such as remote locations or higher transportation costs.																																															
Recommendation	1.Hence , more items of less value can be easily shipped to these people if warehouses are located in these states																																															

	2.States with higher total freight values might represent larger markets or regions with higher demand for Target's products. Understanding the dynamics of these regions can inform market expansion strategies. 3.Freight costs can influence pricing strategies. By analyzing the data, Target can assess whether pricing adjustments are needed in certain regions to optimize profitability while remaining competitive. 4.High freight costs may impact customer satisfaction. Target may want to investigate whether customers in regions with higher freight costs are expressing any concerns or if there are opportunities for improvement in shipping services. 5.States with lower average freight values may indicate areas where Target's logistics operations are relatively more efficient. Identifying and replicating best practices from these regions can enhance overall operational efficiency.
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V. Analysis based on sales, freight and delivery time.

A. Find the no. of days taken to deliver each order from the order's purchase date as delivery time.

Also, calculate the difference (in days) between the estimated & actual delivery date of an order.

Do this in a single query.

Query

```
Select
order_id,
order_purchase_timestamp,
order_estimated_delivery_date,
order_delivered_customer_date,
TIMESTAMP_DIFF(order_delivered_customer_date,order_purchase_timestamp, DAY) as
delivery_time_in_days ,
TIMESTAMP_DIFF(order_estimated_delivery_date,order_delivered_customer_date, DAY) as
diff_estimated_delivery

from `target.orders`
order by 2
```

o/p

Row	order_id	order_purchase_timestamp	order_estimated_delivery_date	order_delivered_customer_date	delivery_time_in_day	diff_estimated_delivery
1	2e7a8482f6fb...	2016-09-04 21:15:19 UTC	2016-10-20 00:00:00 UTC	null	null	null
2	e5fa5a72109...	2016-09-05 00:15:34 UTC	2016-10-28 00:00:00 UTC	null	null	null
3	809a282bbd5...	2016-09-13 15:24:19 UTC	2016-09-30 00:00:00 UTC	null	null	null
4	bfb0f9bdef8...	2016-09-15 12:16:38 UTC	2016-10-04 00:00:00 UTC	2016-11-09 07:47:38 UTC	54	-36
5	71303d7e93b...	2016-10-02 22:07:52 UTC	2016-10-25 00:00:00 UTC	null	null	null
6	3b697a20d9e...	2016-10-03 09:44:50 UTC	2016-10-27 00:00:00 UTC	2016-10-26 14:02:13 UTC	23	0
7	be5bc2f0da1...	2016-10-03 16:56:50 UTC	2016-11-07 00:00:00 UTC	2016-10-27 18:19:38 UTC	24	10
8	65d1e226dfa...	2016-10-03 21:01:41 UTC	2016-11-25 00:00:00 UTC	2016-11-08 10:58:34 UTC	35	16
9	a41c8759fbe...	2016-10-03 21:13:36 UTC	2016-11-29 00:00:00 UTC	2016-11-03 10:58:07 UTC	30	25
10	d207cc27267...	2016-10-03 22:06:03 UTC	2016-11-23 00:00:00 UTC	2016-10-31 11:07:42 UTC	27	22
11	cd3b8574c82...	2016-10-03 22:31:31 UTC	2016-11-23 00:00:00 UTC	2016-10-14 16:08:00 UTC	10	39
12	ae8a60e4b03...	2016-10-03 22:44:10 UTC	2016-12-01 00:00:00 UTC	2016-11-03 14:04:50 UTC	30	27

Observations

1. There are 3 types of evaluated data in columns diff_estimated_delivery :

a) Null values – which shows that order has not been delivered (cross check in order_delivered_customer_date as null)

b) Negative values (e.g highlighted -36) : which shows that order has reached the customer **late : from estimated delivery date**

c) Positive Values : which shows order has reached the customer before the estimated delivery date

Recommendation

Analysing the above data we can say that where there are :

1) **Null Values** : Target needs to look for order_ids with null values in 'diff_estimated_delivery' column and try to find out why has the order not reached the customer – some orders are not delivered since 2016 (considering data available till 2018)

2) Now this could be due to payment not made fully or installments not completed , but then giving customer an estimated delivery date and then not delivering – may lead to huge customer dissatisfaction

3) Also , the parcel may have been lost , there could be many reasons – needs to be investigated

1) **For -ve values :**

1) Giving customer an estimated delivery date and then delivering it late – again huge reason for customer dissatisfaction : Imagine if it was a birthday gift !!

2) Investigation need be done to see what went wrong –

a) are logistics and the logic calculating estimated delivery date in sync or not

b) or the stock is not available

c) or weather conditions were bad for delivery

d) or delivery partner is not diligent enough to give timely delivery etc

1) **Positive values :**

- | | |
|--|--|
| | <ol style="list-style-type: none"> 1) It indicates the item reached customer before delivery date : which is a delightful experience : under promise , over deliver 2) Measures should be taken to repeat such great experiences for customer 3) Study should be done how logistics were managed , was the warehouse nearer , or delivery partner was diligent etc and should try to study it and implement where late deliveries are happening |
|--|--|

B. Find out the top 5 states with the highest & lowest average freight value.

Query	<pre> Select states, avg_freight_value from ((SELECT c.customer_state as states, avg(oi.freight_value) as avg_freight_value from `target.customers` as c join `target.orders` as o on o.customer_id = c.customer_id join `target.order_items` as oi on o.order_id = oi.order_id group by c.customer_state order by avg_freight_value asc limit 5) union all (SELECT c.customer_state as states, avg(oi.freight_value) as avg_freight_value from `target.customers` as c join `target.orders` as o on o.customer_id = c.customer_id join `target.order_items` as oi on o.order_id = oi.order_id </pre>
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	<pre>group by c.customer_state order by avg_freight_value desc limit 5)) order by avg_freight_value asc</pre>																																			
o/p	<table><tr><th>Row</th><th>states</th><th>avg_freight_value</th></tr><tr><td>1</td><td>SP</td><td>15.14727539041...</td></tr><tr><td>2</td><td>PR</td><td>20.53165156794...</td></tr><tr><td>3</td><td>MG</td><td>20.63016680630...</td></tr><tr><td>4</td><td>RJ</td><td>20.96092393168...</td></tr><tr><td>5</td><td>DF</td><td>21.04135494596...</td></tr><tr><td>6</td><td>PI</td><td>39.14797047970...</td></tr><tr><td>7</td><td>AC</td><td>40.07336956521...</td></tr><tr><td>8</td><td>RO</td><td>41.06971223021...</td></tr><tr><td>9</td><td>PB</td><td>42.72380398671...</td></tr><tr><td>10</td><td>RR</td><td>42.98442307692...</td></tr></table>	Row	states	avg_freight_value	1	SP	15.14727539041...	2	PR	20.53165156794...	3	MG	20.63016680630...	4	RJ	20.96092393168...	5	DF	21.04135494596...	6	PI	39.14797047970...	7	AC	40.07336956521...	8	RO	41.06971223021...	9	PB	42.72380398671...	10	RR	42.98442307692...		
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10	RR	42.98442307692...																																		
Summary	1.The first 5 are the ‘bottom 5 states ‘ with lowest freight value 2. The values from row 6-10 are the ‘top_5 states ‘ with highest freight value 3. All of them have been arranged increasing order of the average freight value (as per hint)																																			
Observations	1. States like (SP, PR, MG, RJ,DF) have almost half the avg freight values of that of states like (PI, AC, RO, PB, RR) 2. A generalized observation is that states SP, PR, MG, RJ,DF : a) are major economic and business hubs, being highly populated states, may benefit from economies of scale in logistics b) These states likely have robust transportation infrastructure, including highways, railways, and airports , hence smoother transportation and reduced costs c) SP and RJ are relatively close to major ports, facilitating international shipping potentially reducing inland transportation costs for imported goods.																																			

	d) Higher market competitiveness in these states may lead to competitive pricing among logistics providers and shipping carriers thereby reducing freight cost e) These states are relatively accessible in terms of geographical location , leading to cost effective transportation 3. <u>For states like (PI, AC, RO, PB, RR):</u> a) These states may face geographical challenges such as remoteness, lower population density, or difficult terrain. These factors can contribute to higher transportation costs and longer delivery times. b) Similarly limited Infrastructure, longer transportation distances, limited access to Ports, less market competitiveness all may lead to higher freight values i.e almost double c) Maybe heavy or bulky items are being transported to these states leading to higher freight values
Recommendation	1) <u>Optimize Logistics in High-Cost Regions:</u> For states (PI, AC, RO, PB, RR) Target needs to optimize the freight values by a) negotiating better shipping rates b) establishing regional distribution centers c) looking for local partnerships 2) <u>Evaluate Pricing Strategies:</u> a) Businesses may consider maintaining competitive pricing or even offering promotional shipping incentives to attract more customers. b) Discounts or coupons should be provided for these regions 3) <u>Transparent Communication:</u> Transparent communication about shipping fees can help manage customer expectations and reduce the likelihood of dissatisfaction. 3) <u>Product Weight and Dimensions Optimization:</u> Consider optimizing packaging and product design to reduce shipping costs without compromising product safety. 4) <u>Customer Satisfaction Surveys:</u> Conduct customer satisfaction surveys to gather feedback specifically on the shipping experience and identify problems . 5) <u>Investigate Regional Demand and Supply Chain:</u> Depending upon demand and supply consider setting up warehouses strategically to reduce freight cost . 6) <u>Broader approach :</u> A study can be conducted regarding above points in collaboration with logistics partners and carriers can be crucial in implementing effective changes .

C. Find out the top 5 states with the highest & lowest average delivery time

Query	<pre>SELECT state, delivery_time_in_days FROM (SELECT c.customer_state AS state, AVG(TIMESTAMP_DIFF(o.order_delivered_customer_date, o.order_purchase_timestamp, DAY)) AS delivery_time_in_days, DENSE_RANK() OVER (ORDER BY AVG(TIMESTAMP_DIFF(o.order_delivered_customer_date, o.order_purchase_timestamp, DAY)) ASC) AS ascending_ranking, DENSE_RANK() OVER (ORDER BY AVG(TIMESTAMP_DIFF(o.order_delivered_customer_date, o.order_purchase_timestamp, DAY)) DESC) AS descending_ranking FROM `target.customers` AS c JOIN `target.orders` AS o ON o.customer_id = c.customer_id GROUP BY state) WHERE ascending_ranking BETWEEN 1 AND 5 OR descending_ranking BETWEEN 1 and 5 ORDER BY delivery_time_in_days asc</pre> <pre>SELECT state, delivery_time_in_days FROM (SELECT c.customer_state AS state, AVG(TIMESTAMP_DIFF(o.order_delivered_customer_date, o.order_purchase_timestamp, DAY)) AS delivery_time_in_days, DENSE_RANK() OVER (ORDER BY AVG(TIMESTAMP_DIFF(o.order_delivered_customer_date, o.order_purchase_timestamp, DAY)) ASC) AS ascending_ranking, DENSE_RANK() OVER (ORDER BY AVG(TIMESTAMP_DIFF(o.order_delivered_customer_date, o.order_purchase_timestamp, DAY)) DESC) AS descending_ranking FROM `target.customers` AS c JOIN `target.orders` AS o ON o.customer_id = c.customer_id GROUP BY state) WHERE ascending_ranking BETWEEN 1 AND 5 OR descending_ranking BETWEEN 1 and 5 ORDER BY delivery_time_in_days asc</pre>
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o/p	Row	state	delivery_time_in_days
	1	SP	8.2980614890726656
	2	PR	11.526711354864963
	3	MG	11.543813298106565
	4	DF	12.509134615384614
	5	SC	14.479560191711288
	6	PA	23.316067653276953
	7	AL	24.040302267002509
	8	AM	25.986206896551735
	9	AP	26.731343283582088
	10	RR	28.975609756097562
Observations	1) SP,PR,MG,DF,SC - These states have really low delivery time – i.e fastest delivery in the south , southeastern states 2) PA,AL,AM,AP,RR – these states have almost 3x slower delivery than the above states and tend to fall in the North and North Eastern part of Brazil 3) The fastest states in terms of delivery almost match with states with the lowest freight values which is due to the advantages mentioned in the above 5.2 analysis		
Recommendation	1) For states in north and north eastern part having slower delivery a comparison can be made with Amazon – a indirect competitor to Target– to see what they are using and implement the same 2) Things like : Shipping through drones to difficult geographical regions can be done 3) If Target introduces things like Amazon such as : Amazon Prime: Amazon Prime offers fast and free shipping on eligible items, along with additional benefits such as access to streaming services, exclusive deals, and more. This subscription service encourages customer loyalty and frequent purchases. Amazon Logistics: Amazon has been investing in its logistics network, including a fleet of delivery vans and the use of independent contractors to provide last-mile delivery services. This allows Amazon to have greater control over its delivery operations. Same-Day and One-Day Delivery:		

Amazon has been expanding its same-day and one-day delivery options, providing customers with the convenience of receiving their orders quickly. This initiative aims to meet customer expectations for fast and efficient delivery.

Amazon Locker:

Amazon Prime Air:

Amazon has been exploring the use of drones for package delivery through its Prime Air initiative. While regulatory approvals are still evolving, this technology has the potential to further expedite deliveries.

Hub by Amazon:

Hub by Amazon is a service designed for apartment buildings and residential communities. It provides secure package delivery lockers, making it convenient for residents to receive packages at any time.

Amazon Flex:

Amazon Flex is a program that allows individuals to sign up as independent contractors to deliver Amazon packages using their own vehicles. This model enables flexibility in delivery operations.

Advanced Technology:

Amazon continues to invest in advanced technologies such as artificial intelligence (AI), machine learning, and automation to optimize logistics operations, route planning, and warehouse efficiency.

Global Expansion:

Amazon has been expanding its global footprint, establishing fulfillment centers and distribution hubs in strategic locations to reduce delivery times and costs worldwide.

Customer Feedback and Continuous Improvement:

Amazon actively collects customer feedback and uses it to make improvements to its delivery services. This customer-centric approach helps Amazon adapt to changing preferences and expectations.

D. Find out the top 5 states where the order delivery is really fast as compared to the estimated date of delivery.

You can use the difference between the averages of actual & estimated delivery date to figure out how fast the delivery was for each state.

Query	<pre>SELECT c.customer_state AS state, AVG(TIMESTAMP_DIFF(o.order_delivered_customer_date, o.order_purchase_timestamp, DAY)) AS avg_actual_delivery_days, AVG(TIMESTAMP_DIFF(o.order_estimated_delivery_date, o.order_purchase_timestamp, DAY)) AS avg_estimated_delivery_days,</pre>
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	<pre>(AVG(TIMESTAMP_DIFF(o.order_delivered_customer_date, o.order_purchase_timestamp, DAY)) - AVG(TIMESTAMP_DIFF(o.order_estimated_delivery_date, o.order_purchase_timestamp, DAY))) AS delivery_time_difference FROM `target.customers` AS c JOIN `target.orders` AS o ON o.customer_id = c.customer_id WHERE o.order_status = 'delivered' GROUP BY state ORDER BY delivery_time_difference ASC LIMIT 5</pre> <pre>SELECT c.customer_state AS state, AVG(TIMESTAMP_DIFF(o.order_delivered_customer_date, o.order_purchase_timestamp, DAY)) AS avg_actual_delivery_days, AVG(TIMESTAMP_DIFF(o.order_estimated_delivery_date, o.order_purchase_timestamp, DAY)) AS avg_estimated_delivery_days, (AVG(TIMESTAMP_DIFF(o.order_delivered_customer_date, o.order_purchase_timestamp, DAY)) - AVG(TIMESTAMP_DIFF(o.order_estimated_delivery_date, o. order_purchase_timestamp, DAY))) AS delivery_time_difference FROM `target.customers` AS c JOIN `target.orders` AS o ON o.customer_id = c.customer_id WHERE o.order_status = 'delivered' GROUP BY state ORDER BY delivery_time_difference ASC LIMIT 5;</pre>																														
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5	RR	28.97560975609...	45.63414634146...	-16.658536585365852																											
Observations	1) The delivery is really fast in these states : AC,RO,AP,AM,RR states where it is the fastest in AC i.e on an average 20 days earlier than the estimated delivery date , RO – 2 nd fastest and so on																														

	2) The average time for approval is 0.27 days <pre>Select round(Avg(TIMESTAMP_DIFF(o.order_approved_at, o.order_purchase_timestamp, DAY)),2) as approval_time from `target.orders` as o</pre> 3) The average time for carrier to start the delivery itself takes atleast 2.32 days <pre>Select round(Avg(TIMESTAMP_DIFF(o.order_delivered_carrier_date,o.order_approved_at, DAY)),2) as approv_to_carrier from `target.orders` as o</pre>
Recommendation	1) As these are northern states and in the 5.c observation these states have actually slower delivery times – Hence a buffer is kept before giving an estimated delivery time . This practice is good but measures are required to be taken to actually make delivery faster in these states . 2) A study can be made what actually leads to faster delivery and implemented for other states also . 3) The average time for order approval is 0.27 days and the average time for the carrier to start delivery is 2.32 days which can be optimized for better and faster delivery 4) Recommendations would be the same as 5.c

VI. Analysis based on the payments:

A. Find the month on month no. of orders placed using different payment types.

Query	a) <u>If only month on month asked :</u> <pre>SELECT Extract(Month from o.order_purchase_timestamp) as Month_, p.payment_type , count(distinct p.order_id)as no_of_orders from `target.orders` as o left join `target.payments` as p on o.order_id = p.order_id group by Month_,payment_type order by 1,3 desc</pre>
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	b) If month on month – <u>Yearwise</u> data required <pre>SELECT Extract(Year from o.order_purchase_timestamp) as Year_, Extract(Month from o.order_purchase_timestamp) as Month_, p.payment_type, count(distinct p.order_id)as no_of_orders from `target.orders` as o left join `target.payments` as p on o.order_id = p.order_id group by 1,2,3 order by 1,2 asc,4 desc</pre>																																												
o/p	a) If only month on month asked : <table><tr><th>Row</th><th>Month_</th><th>payment_type</th><th>no_of_orders</th></tr><tr><td>1</td><td>1</td><td>credit_card</td><td>6093</td></tr><tr><td>2</td><td>1</td><td>UPI</td><td>1715</td></tr><tr><td>3</td><td>1</td><td>voucher</td><td>337</td></tr><tr><td>4</td><td>1</td><td>debit_card</td><td>118</td></tr><tr><td>5</td><td>2</td><td>credit_card</td><td>6582</td></tr><tr><td>6</td><td>2</td><td>UPI</td><td>1723</td></tr><tr><td>7</td><td>2</td><td>voucher</td><td>288</td></tr><tr><td>8</td><td>2</td><td>debit_card</td><td>82</td></tr><tr><td>9</td><td>3</td><td>credit_card</td><td>7682</td></tr><tr><td>10</td><td>3</td><td>UPI</td><td>1942</td></tr></table>	Row	Month_	payment_type	no_of_orders	1	1	credit_card	6093	2	1	UPI	1715	3	1	voucher	337	4	1	debit_card	118	5	2	credit_card	6582	6	2	UPI	1723	7	2	voucher	288	8	2	debit_card	82	9	3	credit_card	7682	10	3	UPI	1942
Row	Month_	payment_type	no_of_orders																																										
1	1	credit_card	6093																																										
2	1	UPI	1715																																										
3	1	voucher	337																																										
4	1	debit_card	118																																										
5	2	credit_card	6582																																										
6	2	UPI	1723																																										
7	2	voucher	288																																										
8	2	debit_card	82																																										
9	3	credit_card	7682																																										
10	3	UPI	1942																																										

b) If month on month – Yearwise data required

Row	Year_	Month_	payment_type	no_of_orders
1	2016	9	credit_card	3
2	2016	9	null	0
3	2016	10	credit_card	253
4	2016	10	UPI	63
5	2016	10	voucher	11
6	2016	10	debit_card	2
7	2016	12	credit_card	1
8	2017	1	credit_card	582
9	2017	1	UPI	197
10	2017	1	voucher	33
11	2017	1	debit_card	9
12	2017	2	credit_card	1347

Observations

1. Its been observed that for every month
 - a) Credit card is the most popular mode of payment followed by UPI
 - b) Vouchers and debit card are relatively less popular
2. Two discrepancies where the payment type is **null , not defined** has been identified and needs to be investigated
3. Month-wise increase has been seen in credit card payments purchases from 1 to 5 month , 5 then has stabilized , also 8th month being 2nd highest then a sudden decrease is seen It again rises but not to the peak in the months of Nov,Dec
4. UPI seems to follow almost the same path but the highest no of payments made using UPI doesn't go beyond 2100 , average hovers around 1500-2000 and low being around 1000
5. Debit card needs instant money payment with full amount – hence becomes the least favorite amongst consumers

Recommendation

1. Since people prefer credits cards the most , they should be awarded loyalty points or schemes like this for encouraging more sales
2. From business point of view if direct full payments the cash flow will improve , hence UPI and debit card payments should be incentivized using discounts, cashback, or loyalty points for these specific payment types.
3. Something like Amazon Prime – loyalty program can be run to get retain customers and increase repeat sales
4. Something like Buy Now and Pay later kind of schemes can be run making it easier for middle class and lower middle class to make bigger purchases
5. Festive season should provide more modes of payment and easier processing for smoother transactions
6. Collaborate with payment providers to explore opportunities for mutually beneficial promotions or partnerships

	7. Something like Wallets , international modes like Paypal , accepting Bitcoins etc also can be considered 8. Continuously monitor the performance of each payment type and adjust strategies based on changing trends. 9. Emphasize the security measures in place for each payment type to build trust among customers. Ensuring a secure and reliable payment experience can positively impact customer satisfaction and loyalty. 10. Things like zero cost EMI combined with vouchers etc can be introduced to attract wider range of audience driving more sales .
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B. Find the no. of orders placed on the basis of the payment installments that have been paid.

Query	<pre> SELECT count(distinct order_id) as no_of_orders, from `target.payments` where payment_installments >= 1 </pre>
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o/p

Row	no_of_orders
1	99438

But there are 3 orders where payment installment is 1 but payment value is 0

Row	order_id	payment_sequential	payment_type	payment_installment	payment_value
1	00b1cb0320190ca0daa2c88b3...	1	not_defined	1	0.0
2	4637ca194b6387e2d538dc89b...	1	not_defined	1	0.0
3	c8c528189310eaa44a745b8d9...	1	not_defined	1	0.0

Hence , if you want to eliminate such orders :

```
SELECT
    count( distinct order_id) as no_of_orders,
from
    `target.payments`
where payment_installments >= 1 and payment_value > 0
```

Row	no_of_orders
1	99435

Observations

1. In all there are 99438 orders where at least 1 payment installments is made . However , upon further investigation it was seen that there are 3 orders with order_ids :

"00b1cb0320190ca0daa2c88b35206009"
"4637ca194b6387e2d538dc89b124b0ee"
"c8c528189310eaa44a745b8d9d26908b"

Which have payment method as 'not defined' , payment installment =1 but payment value is zero as shown in the output .

If you eliminate these 3 orders then the ideal :

Count of orders placed on the basis of the payment installments that have been paid = 99435

2. The fact that a significant number of orders involve payment installments suggests that installment payment options are popular among customers.
3. People don't want to pay upfront total amount and divide it into installments giving them scope financial planning

	4. People with lesser income can buy high priced item using installments thereby increasing the purchasing power resulting into more sales for Target 5. Successful installment payments indicate a positive customer experience with the chosen payment method. The ability to complete installments successfully suggests that customers are comfortable and satisfied with this payment option.
Recommendation	1. Target needs to investigate the orders with order_ids : "00b1cb0320190ca0daa2c88b35206009" "4637ca194b6387e2d538dc89b124b0ee" "c8c528189310eaa44a745b8d9d26908b" If these are legit orders or not or some kind of scam or a free or discounted installment or a bug in the system because it has captured the payment installment as 1 , however the payment value is zero. 2. Provide additional benefits like convertible loyalty points and discounts , exclusive deals , vouchers for people for paying timely installments incentivizing continued use of installment options. 3. Monitor Default Rates :Keep a close eye on default rates for installment payments. If there is an increase in defaults, it may indicate a need to reassess creditworthiness checks . 4. Collaborate with payment providers to explore innovative installment solutions or partnerships . This can benefit customers as well as Target in monetary sense – a win win situation .